

SEM-II

Course: MTMDSC202T

Calculus (Marks: 100, Credits: 5)

Unit – 1 : Limits, Continuity and Differentiability

Limit of a function, ϵ - δ definition of a limit, Infinite limits, Continuity and types of discontinuities; Differentiability of a function, Relation between differentiability and continuity, Successive differentiation, Leibnitz theorem and its applications to functions such as $e^{ax+b} \sin x$, $e^{ax+b} \cos x$, $(ax+b)^n \sin x$, $(ax+b)^n \cos x$; Partial differentiation, Euler's theorem on homogeneous functions and its converse.

Unit – 2 : Mean Value Theorems and its Applications

Rolle's theorem, Lagrange's mean value theorem, Cauchy's mean value theorem, Geometrical interpretation of mean value theorems and applications to monotonic functions and inequalities; Taylor's theorem, Taylor's series, Maclaurin's series expansions of e^x , $\sin x$, $\cos x$, $\log(1+x)$, $(1+x)^m$; Indeterminate forms.

Unit -3 : Integral Calculus

Integration of rational and irrational functions, Evaluation of definite integrals, Special integrals, Reduction formulae, derivations and illustrations of reduction formulae for the integration of $\sin^n x$, $\cos^n x$, $\tan^n x$, $\sec^n x$, $(\log x)^n$, $\sin^n x \cos^m x$, $e^{ax} \cos^n x$, $(x^2 + a^2)^n$ and their applications; Improper integrals, Beta and Gamma functions.

Unit – 4 : Applications

Tangent and Normal; Curvature; Asymptotes of general algebraic curves, Parallel asymptotes, Asymptotes parallel to axes; Envelopes; Maxima and Minima; Concavity and convexity, Points of inflexion; Tracing of Cartesian and polar curves; Length of plane curve and area bounded by plane curves, Volume and surface area of solids of revolution.

Graphical Demonstration (Teaching Aid)

1. Plotting of graphs of function e^{ax+b} , $\log(ax+b)$, $1/(ax+b)$, $\sin(ax+b)$, $\cos(ax+b)$, $|ax+b|$ and to illustrate the effect of a and b on the graph.
2. Plotting the graphs of polynomial of degree 4 and 5, the derivative graph, the second derivative graph and comparing them.

3. *Sketching parametric curves.*
4. *Obtaining volume and surface of revolution of curves.*
5. *Tracing of conics in Cartesian coordinates/polar coordinates.*

Books Recommended:

- G.B. Thomas and R.L. Finney, Calculus, 9th Ed., Pearson Education, Delhi, 2005.
- M.J. Strauss, G.L. Bradley and K. J. Smith, Calculus, 3rd Ed., Dorling Kindersley (India) P. Ltd. (Pearson Education), Delhi, 2007.
- Gorakh Prasad, Differential Calculus (19th edition), Pothishala Pvt. Ltd., 2016.
- R. Courant and F. John, Introduction to Calculus and Analysis (Volumes I & II), Springer-Verlag, New York, Inc., 1989.
- Gorakh Prasad, Integral Calculus, Pothishala Pvt. Ltd., Allahabad, 2015.
- Gabriel Klambauer, Aspects of Calculus, Springer-Verlag, 1986.
- Howard Anton, I. Bivens& Stephan Davis, Calculus (10th edition), Wiley India, 2016.
- T. Apostol, Calculus, Volumes I and II.
- S. Goldberg, Calculus and Mathematical analysis.

Question pattern for End Semester Examination:

- (i) Short answer type questions (10 marks): FIVE questions out of EIGHT questions of 2 marks each to be answered.
- (ii) Broad questions (40 marks): EIGHT questions of 8 marks each (which may be subdivided into multiple parts) will be set taking two from each unit. FIVE questions out of EIGHT are to be answered.